

Lab 1 – Lab Reflection

SRN: PES1UG24AM221

System: Domain & SSL Certificate Expiry Alert System

1. What I Learned

In this lab, I learned how to identify, analyse, and document software requirements for a system. I understood the difference between functional and non-functional requirements and learned how system requirements can be represented using UML use-case modelling. I also learned how actors, use cases, and relationships such as <<include>> and <<extend>> are used to describe system interactions.

2. Requirements Engineering Experience

For the Domain & SSL Certificate Expiry Alert System, I identified the major requirements related to domain monitoring, SSL/TLS certificate monitoring, expiration detection, alerts, auditing, and secure access. I documented the requirements in a structured form so that they could be used as a foundation for later design and development.

3. UML Use-Case Modelling Experience

I identified the main actors as the SysAdmin and Security Officer and modelled their interactions with the system. The use-case diagram helped me understand how users interact with system functions and how related use cases can be connected through include and extend relationships. I also documented the main success scenario and an alternate flow for audit failure.

4. Challenges Faced

One challenge was deciding the correct level of detail for the requirements and use cases. It was important to avoid making the diagram too complicated while still representing the important system behaviour. I also had to ensure that the relationships between use cases were logically represented.

5. Skills Developed

This lab improved my skills in requirements elicitation, functional and non-functional requirement documentation, actor identification, use-case analysis, UML modelling, and writing structured use-case flows. I also gained experience in translating a system idea into clear and testable software requirements.

6. Overall Reflection

Overall, this lab helped me understand that clear requirements are essential before designing or implementing a software system. Creating the use-case model made the system easier to understand from the users' perspective and provided a strong foundation for the architectural and Agile activities carried out in the following labs.